

Financing has to be a strategic mix of public and private resources. Green bonds and climate-friendly investment strategies can attract private-sector funds. Carbon capture and storage (CCS) pilot projects can provide solutions for industries that are facing challenges in decarbonising.

But India's renewable energy story cannot be written by only the Union government: the states must work in tandem.

Anish De, consultancy firm KPMG's Global Head for Energy Natural Resources & Chemicals, says India has built a strong enabling environment with favourable policies, regulations, and proactive infrastructure that includes transmission, renewable parks, and demand aggregation. "There is still a challenge at the state level...state policies and regulations are not always favourable and are also inconsistent at times," says De. The states could provide a massive fillip to the use of renewable energy, from grid-scale to rooftop applications.

DECARBONISING INDUSTRIES

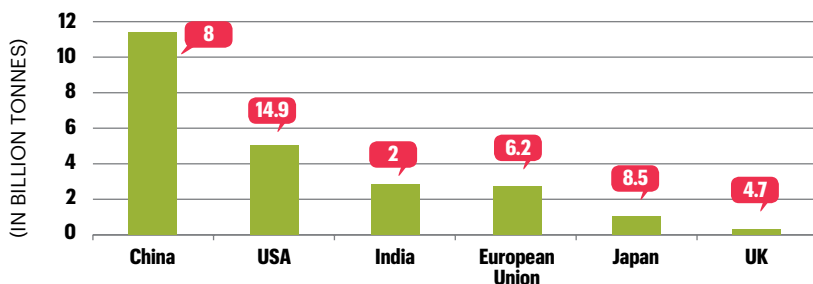
Industry accounts for around 28% of total greenhouse gas (GHG) emissions, most coming from metals, minerals, manufacturing, and rubber & plastic. All these rely heavily on electricity generated from fossil fuels. What could be done for these sub-sectors over the next five years?

One way is to develop and adopt automation and energy management systems in industries such as cement, steel, construction and textiles. Steel, cement, and fertilisers can move towards power from green hydrogen, produced by splitting water molecules using solar power. The hydrogen is then made to react with oxygen across an electrochemical cell to generate electricity.



A GREY AREA

- China is the biggest GHG emitter globally, followed by the US and India
- However, India's per capita emissions are much lower than in most other countries

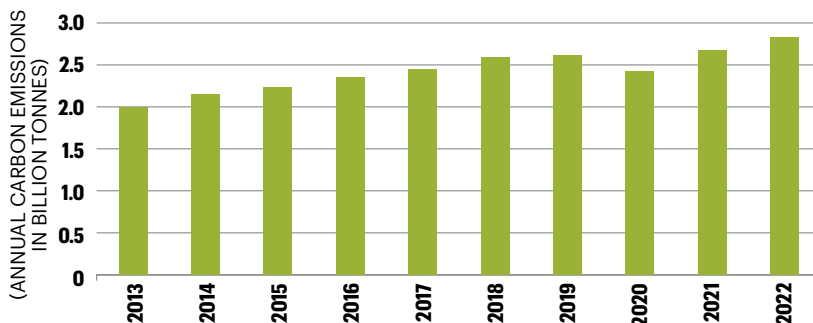


● ANNUAL CO₂ EMISSIONS ● PER CAPITA EMISSIONS (IN TONNES)

SOURCE GLOBAL CARBON BUDGET 2023

INDIA'S CARBON EMISSIONS

- India is a developing economy with most of the urban infrastructure yet to be built; its carbon emissions are on the rise and will peak in the next two decades



SOURCE GLOBAL CARBON BUDGET 2023